



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
12-07-2020

JEE(Main + Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II, III, IV(A) & IV]

PAPER-1

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

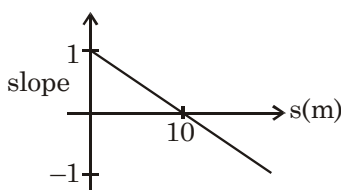
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks: 28)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

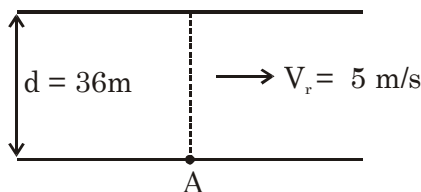
1. A projectile is fired on a horizontal ground. Variation of slope of its trajectory with horizontal displacement is shown ($g = 10 \text{ m/s}^2$). Choose the **CORRECT** statement(s) :



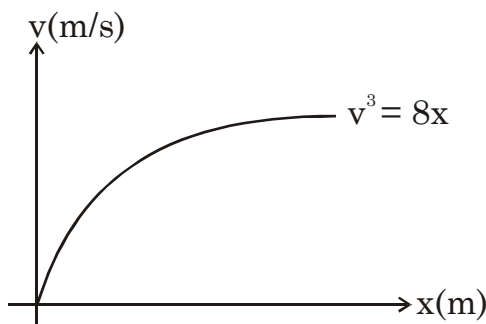
- (A) Angle of projection of projectile is 60°
 (B) Horizontal range of projectile is 20 m
 (C) Time of flight of projectile is 2sec.
 (D) Area under curve gives vertical displacement of projectile.

Space for Rough Work

2. A boat starts from one bank of the river from point A to reach other by crossing width d of the river with minimum drift. Speed of a boat with respect to river is 4 m/s and speed of river with respect to ground is 5 m/s . Value of $d = 36 \text{ m}$, then



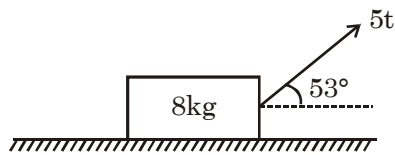
- (A) Minimum drift will be zero.
 (B) Minimum drift will be 27 m .
 (C) Time taken for minimum drift will be 15 sec .
 (D) Angle at which boat will be steered from river flow direction for minimum drift is 127° .
3. The v - x curve for an object moving along the x -axis shown in the diagram :



- (A) The acceleration of particle when $x = 1 \text{ m}$ is $\left(\frac{4}{3}\right) \text{ m/s}^2$
 (B) The acceleration of particle when $x = 8 \text{ m}$ is $\left(\frac{2}{3}\right) \text{ m/s}^2$
 (C) The acceleration of particle when $v = 2 \text{ m/s}$ is $\left(\frac{8}{3}\right) \text{ m/s}^2$
 (D) The acceleration of particle when $v = 8 \text{ m/s}$ is $\left(\frac{1}{3}\right) \text{ m/s}^2$

Space for Rough Work

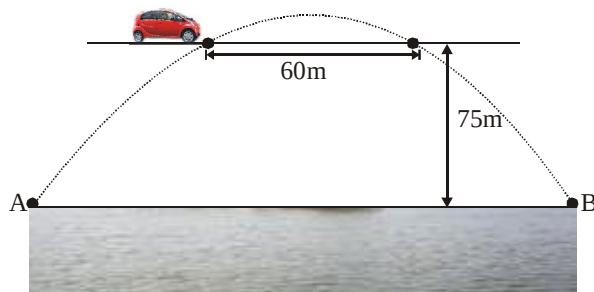
4. A block of mass $m = 8 \text{ kg}$ is kept on ground. A force of " $5t$ " Newton acts at an angle of 53° with horizontal, where t is time in second starts acting on it at $t = 0$ as shown in figure. All surfaces are frictionless.



- (A) Acceleration of block at $t = 40 \text{ sec}$ is $(15\hat{i} + 20\hat{j}) \text{ m/s}^2$
- (B) Acceleration of block at $t = 40 \text{ sec}$ is $(15\hat{i} + 10\hat{j}) \text{ m/s}^2$
- (C) Velocity of block at $t = 40 \text{ sec}$ is $(300\hat{i} + 100\hat{j}) \text{ m/s}$
- (D) Velocity of block at $t = 40 \text{ sec}$ is $(300\hat{i} + 180\hat{j}) \text{ m/s}$

Space for Rough Work

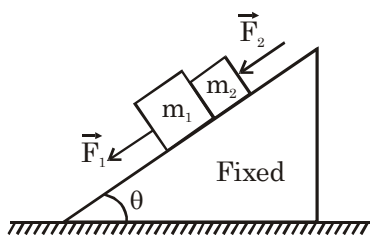
5. In the figure shown there is a long horizontal bridge over a river 75 m high from water surface. A strong man throws a stone in the parallel plane of the bridge. A observer in a car travelling on the bridge finds the stone going pass by the car while ascending and also while descending between two points on the road 60 m away. The car is travelling at a speed of 30 m/s. The stone is thrown from the bank of river just at the same level of water, then



- (A) Angle of projection is 37° with horizontal direction.
 (B) Time of flight of stone is 8 sec.
 (C) Range(AB) of stone is 240 m
 (D) Angle made by velocity vector with horizontal direction at 1 sec after projection is 45° .

Space for Rough Work

6. The position vector of a particle varies with time as $\vec{r} = (\cos t)\hat{i} + (\sin t)\hat{j} + (3t^2 + 1)\hat{k}$. The particle :
- (A) Has constant acceleration.
 (B) Speed increases
 (C) Velocity and acceleration are orthogonal at $t = 0$
 (D) Has variable acceleration.
7. Two blocks of masses m_1 and m_2 ($m_1 > m_2$) are placed on the fixed inclined plane. Forces of $\vec{F}_1$ and $\vec{F}_2$ are applied on blocks m_1 and m_2 respectively along the inclined plane in downward direction as shown in figure. Here 'N' is the normal contact force between the blocks. The surface between blocks and inclined plane is smooth. Select the **CORRECT** alternative.



- (A) $N = 0$ if $\vec{F}_1 = \vec{F}_2 = 0$ (B) $N = 0$ if $\vec{F}_1 = \vec{F}_2 \neq 0$
- (C) $N \neq 0$ if $\vec{F}_1 = \vec{F}_2 \neq 0$ (D) $N = 0$ if $\frac{\vec{F}_1}{m_1} = \frac{\vec{F}_2}{m_2}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

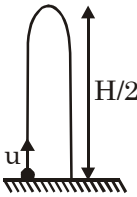
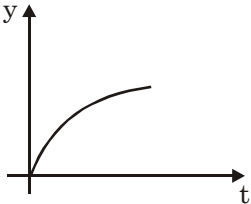
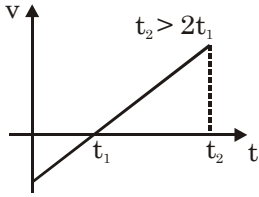
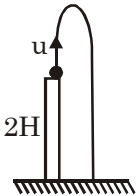
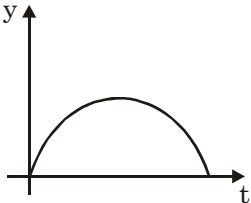
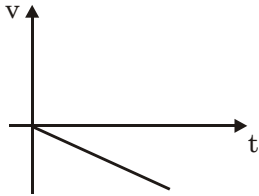
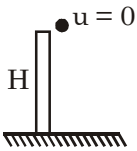
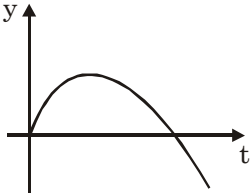
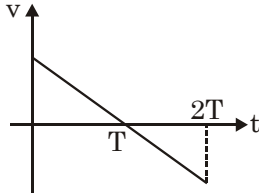
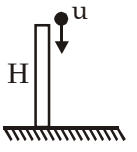
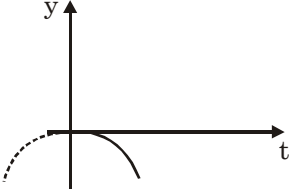
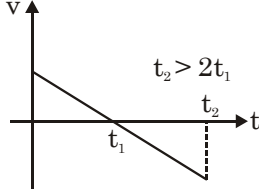
- This section contains **SIX** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

Column-1 represents different cases of vertical motion under gravity, with point of projection (or drop) as origin and upward (y) direction as positive.

Column-2 represents position-time (y-t) graph of motion

Column-3 represents velocity-time (v-t) graph of motion

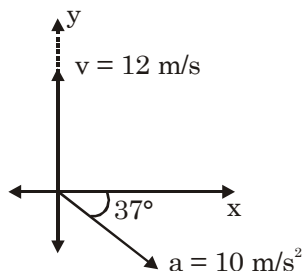
Column-1	Column-2	Column-3
(I) 	(i) 	(P) 
(II) 	(ii) 	(Q) 
(III) 	(iii) 	(R) 
(IV) 	(iv) 	(S) 

8. In which case particle is speeding down then speeding up for equal time intervals
(A) (II) (ii) (R) (B) (I) (ii) (R) (C) (II) (iii) (R) (D) (I) (iii) (S)
9. In which case particle strikes the ground with maximum speed
(A) (II) (i) (S) (B) (IV) (iii) (R) (C) (IV) (i) (R) (D) (II) (iii) (S)
10. In which case distance travelled by particle is minimum and speed of striking the ground is not minimum of given cases.
(A) (III) (iv) (Q) (B) (I) (iv) (Q) (C) (IV) (iii) (R) (D) (IV) (iv) (R)

Space for Rough Work

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given in the three columns of the following table.

Diagram shows velocity and acceleration of a particle, initially at origin at $t = 0$.



Column-1 x-coordinate	Column-2 y-coordinate	Column-3 Speed (in m/s)
(I) 0 m	(i) 0 m	(P) 10
(II) 4 m	(ii) 6 m	(Q) 16
(III) 16 m	(iii) 9 m	(R) 32
(IV) 64 m	(iv) 12 m	(S) $4\sqrt{73}$
11. At $t = 1$ sec :-		
(A) (II) (iv) (P)	(B) (I) (III) (Q)	(C) (II) (iii) (P)
		(D) (III) (IV) (Q)
12. When y component of velocity is zero :-		
(A) (II) (iv) (Q)	(B) (III) (iv) (Q)	(C) (IV) (iii) (R)
		(D) (I) (iii) (Q)
13. When $v_y = -12$ m/s :-		
(A) (IV) (i) (S)	(B) (I) (i) (Q)	(C) (IV) (iv) (S)
		(D) (I) (ii) (R)

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	●	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	●	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

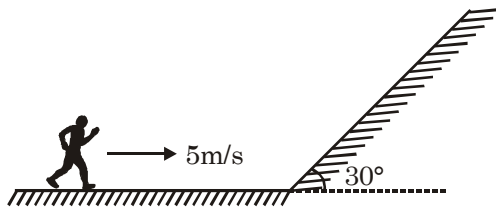
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Velocity of a particle moving on straight line varies as $v = \frac{2}{x}$; where $x > 0$ and at $t = 1$ sec,

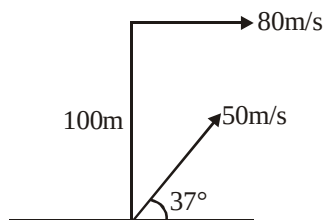
$x = 2$ m. Its velocity at $t = 4$ sec is found $\frac{n}{4}$ m/s. Find n .

Space for Rough Work

2. A man was running on horizontal road with speed 5 m/sec observed rain is falling vertically with speed $5\sqrt{3}$ m/s. Now he is running on inclined plane of inclination 30° with same speed 5 m/sec, then he observed rain is falling with speed $5\sqrt{x}$. Find value of x.

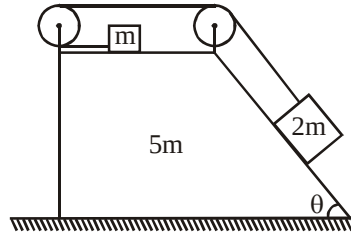


3. A particle is thrown horizontally with a speed of 80 m/s from a 100 m high building. Simultaneously another particle is projected from the foot of building with a speed of 50 m/s at an angle of 37° with the horizontal. The minimum distance between the two particles is $10K$, find value of K.

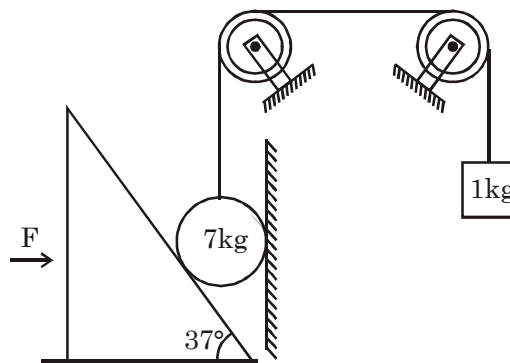


Space for Rough Work

4. In the system shown the acceleration of the wedge of mass $5m$ is zero if angle ' θ ' of wedge is $\frac{\pi}{x}$, then calculate x (all surfaces are smooth).



5. The situation shown in the figure, all surfaces are smooth. The mass of sphere is 7kg . If the force F required to keep the system in equilibrium is $9n\text{ N}$. Find the value of n .



Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 28)**

- This section contains **SEVEN** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
1. An element 'X' having atomic number '25' on removal of two electrons, changes into its dipositive cation 'Y'. Which of the following statement/s is/are **INCORRECT** regarding 'X' and 'Y'.
(A) spin only magnetic moment : $X > Y$
(B) effective nuclear charge on outermost electron : $X > Y$
(C) Total number of d-electrons : $X = Y$
(D) Radius : $X < Y$
 2. Which of the following is/are **INCORRECT** order of radius.
(A) $\text{Ca}^{2+} < \text{S}^{2-} < \text{Cl}^-$ (B) $\text{N} < \text{Li} < \text{B}$
(C) $\text{Na}^+ < \text{Mg}^{2+} < \text{Cl}^-$ (D) $\text{Cl} < \text{Mg} < \text{Ca}$

Space for Rough Work

3. **CORRECT** statement/s for NO_3^- ion is/are :
- (A) $\angle \text{ONO} = 120^\circ$
 (B) Nodal plane of $(2p_\pi - 2p_\pi)$ bond lies in molecular plane of ion
 (C) Covalency of N-atom is five
 (D) Bond order of N-O bond is 1.33
4. Which properties is/are belongs to element with atomic number 17.
 (A) Its electron affinity is highest among all known elements
 (B) Its ionisation energy is 2nd highest among all known elements
 (C) Its oxide is acidic in nature
 (D) Its electronegativity is highest among all known elements
5. Which of the following is/are **INCORRECT** order of bond angle?
 (A) $\text{NF}_3 > \text{NH}_3$ (B) $\text{OF}_2 > \text{OH}_2$ (C) $\text{CH}_4 = \text{CF}_4$ (D) $\text{BF}_3 = \text{BCl}_3$
6. If internuclear axis is Z axis, then which set/s of orbital can form π -bond.
 (A) $p_x + p_x$ (B) $p_x + d_{xy}$ (C) $p_y + d_{xy}$ (D) $d_{xz} + d_{xz}$
7. Which molecule(s)/ion(s) can accomodate all oxygen atom present in one plane.
 (A) CO_3^{2-} (B) SO_3^{2-} (C) NO_3^- (D) SO_3

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

- This section contains **SIX** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table.

Column-I (Molecular species)	Column-II (Hybridisation)	Column-III (Property)
(I) XeO_6^{4-}	(i) sp^3	(P) Planar
(II) ClF_3	(ii) sp^3d	(Q) Non planar
(III) I_3^+	(iii) sp^3d^2	(R) Polar
(IV) XeF_4	(iv) sp^3d^3	(S) Non polar

8. Which one of the following option is **CORRECT** combination?
 (A) (I, ii, Q) (B) (II, ii, Q) (C) (I, iii, S) (D) (II, i, R)
9. Which one of the following option is **CORRECT** combination?
 (A) (III, i, R) (B) (IV, iv, P) (C) (III, ii, P) (D) (IV, iii, R)
10. Which are of the following option is **INCORRECT** combination?
 (A) (I, iii, Q) (B) (III, i, P) (C) (II, ii, R) (D) (IV, iv, P)

Space for Rough Work

Compound	Nature of Oxide	Characteristics
(I) Cl_2O_6	(i) Neutral	(P) Reacts with HCl
(II) Al_2O_3	(ii) Basic	(Q) Reacts with NaOH
(III) CaO	(iii) Acidic	(R) Covalent Compound
(IV) CO_2	(iv) Amphoteric	(S) Ionic Compound

11. **CORRECT** set of combination is:-

- (A) (II) (ii) R (B) (I) (iii) Q (C) (III) (i) P (D) (IV) (ii) (R)

12. **INCORRECT** set of combination is:-

- (A) (I) (iii) (R) (B) (II) (iv) (Q) (C) (III) (ii) (R) (D) (IV) (iii) (Q)

13. **CORRECT** set is:-

- (A) (II) (iii) (R) (B) (II) (iv) (R) (C) (III) (iv) (P) (D) (IV) (iii) (Q)

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Among given elements, find total number of elements which having electron affinity greater than oxygen.
 S, Se, Te, N, P, F, Cl, C, Si

Space for Rough Work

2. Find total number of species in which number of lone pair(s) on central atom is less than total number of lone pairs on surrounding atoms

CH_4 , SiH_4 , NF_3 , PF_3 , BrF_3 , SF_2 , H_2O

3. For given oxides calculate $\frac{(X+Y)}{2}$.

FeO , Cr_2O_3 , MnO , ZnO , CO , Na_2O , Al_2O_3 , PbO_2 , BaO

where :

X = Number of oxides which reacts only with acid

Y = Number of neutral oxides

Space for Rough Work

4. For given molecular species calculate $\frac{X}{Y}$

XeF_4 , ClO_4^- , XeO_3 , IF_4^+ , ClF_3 , SF_6 , SF_4 , PCl_6^-

where

X = Number of molecular species in which all bonds having same bond length.

Y = Number of molecular species in which all bonds does not have same bond length.

5. Find sum of Sulphur-oxygen bond order in SO_4^{2-} , SO_3^{2-} and SO_2 .

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 28)**

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following hold(s) good ?

$$(A) \sin 7^\circ \cos 23^\circ + \cos 7^\circ \sin 23^\circ = \frac{1}{2} \quad (B) \sin 41^\circ \cos 11^\circ - \sin 11^\circ \cos 41^\circ = \frac{1}{2}$$

$$(C) \cos 63^\circ \cos 3^\circ - \sin 63^\circ \sin 3^\circ = \frac{1}{2} \quad (D) \cos 49^\circ \cos 4^\circ + \sin 49^\circ \sin 4^\circ = \frac{1}{\sqrt{2}}$$

2. Which of the following statements are correct ?

$$(A) \sin 1 > \sin 1^\circ \quad (B) \tan 2 < 0 \quad (C) \tan 1 > \tan 2 \quad (D) \tan 2 < \tan 1 < 0$$

Space for Rough Work

3. Let $f(n) = \cos 1 \times \cos 2 \times \cos 3 \times \dots \times \cos n \forall n \in \mathbb{N}$ then the number of elements in the set $A = \{f(1), f(2), \dots, f(7)\}$ that are negative are λ then λ is greater than or equal to
 (A) 3 (B) 4 (C) 5 (D) 6
4. For the series $S = 1 + \frac{1}{(1+3)}(1+2)^2 + \frac{1}{(1+3+5)}(1+2+3)^2 + \frac{1}{(1+3+5+7)}(1+2+3+4)^2 + \dots$ then which of the following options is/are correct ?
 (A) T_7 is 16 (B) T_7 is 18 (C) $S_{10} = \frac{505}{4}$ (D) $S_{10} = \frac{405}{4}$
 (When T_n, S_n denotes the n^{th} term and sum of n terms respectively)
5. If $(\log_{\beta} \alpha)^2 + (\log_{\alpha} \beta)^2 = 142$, then value of $\log_{\beta} \alpha + \log_{\alpha} \beta$ can be
 (A) 14 (B) 12 (C) -14 (D) -12
6. If a, b and c are positive real numbers such that $a^{\log_{11} 5} = 121$; $b^{\log_3 49} = \sqrt{3}$ and $c^{\log_7 25} = \sqrt{7}$, then
 (A) $a^{(\log_{11} 5)^2} = 25$ (B) $b^{(\log_3 49)^2} = 49$ (C) $c^{(\log_7 25)^2} = 25$ (D) $a^{(\log_{11} 5)^2} - 5 \cdot c^{(\log_7 25)^2} = 0$
7. If a, b, c are three distinct numbers in G.P., b, c, a are in A.P. and a, bc, abc are in H.P. then the possible value of b is/are
 (A) $3 + 4\sqrt{2}$ (B) $4 + 3\sqrt{2}$ (C) $3 - 4\sqrt{2}$ (D) $4 - 3\sqrt{2}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

- This section contains **SIX** questions of matching type.
- This section contains **Two** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Answer Q.8, Q.9 and Q.10 by appropriately matching the information given in the three columns of the following table

Let $f(x) = x^2 - (m - 3)x + m$,
 $g(x) = \log_2(x^2 - 8)$ and consider

$$t_r = \frac{1}{r(r+1)}, \text{ where } r \in \mathbb{N}$$

- Column I, contains information about $f(x)$
- Column II, contains inequalities on $g(x)$
- Column III, contains sum of sequence $< t_r >$

Column I

Column II

Column III

(I) For $m \in (-\infty, 0)$ exactly one root of $f(x)$ lies in $(0, 2)$	(i) $g(x) > 0$ $\Rightarrow x \in (-\infty, -3) \cup (3, \infty)$	(P) $\sum_{r=1}^{10} t_r = \frac{10}{11}$
(II) For $m \in (0, 10)$ both roots of $f(x)$ lie in $(0, 2)$	(ii) $g(x) < 0$ $\Rightarrow x \in (-3, -2\sqrt{2}) \cup (2\sqrt{2}, 3)$	(Q) $\sum_{r=1}^{10} t_r < \sum_{r=6}^{15} t_r$
(III) For $m \in [0, 1]$ both roots of $f(x)$ do not lie in $(0, 2)$	(iii) $g(x) < 3$ $\Rightarrow x \in (-4, 4)$	(R) $\sum_{r=1}^{\infty} t_r = 1$
(IV) For $m \in (-\infty, 0)$ roots of $f(x)$ are opposite in sign	(iv) $g(x) \geq 3$ $\Rightarrow x \in (-\infty, -4] \cup [4, \infty)$	(S) $\sum_{r=2}^{\infty} t_r = \frac{1}{2}$

8. Which of the following options is the only **CORRECT** combination ?
 (A) (I) (ii) (Q) (B) (II) (iii) (Q) (C) (I) (i) (P) (D) (I) (iii) (R)
9. Which of the following options is the only **CORRECT** combination ?
 (A) (II) (ii) (Q) (B) (III) (ii) (P) (C) (IV) (iii) (Q) (D) (II) (i) (P)
10. Which of the following options is the only **CORRECT** combination ?
 (A) (III) (iii) (R) (B) (IV) (iii) (P) (C) (III) (i) (Q) (D) (IV) (iv) (S)

Space for Rough Work

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given in the three columns of the following table

- $a_1, a_2, a_3, \dots, a_n \dots$ are in A.P.
- $b_1, b_2, b_3, \dots, b_n \dots$ are in G.P.
- and $c_1, c_2, c_3, \dots, c_n \dots$ are in H.P.

Column I	Column II	Column III
(I) If $a_i > 0 \forall i \in \mathbb{N}$ then $\sum_{i=1}^{2020} \frac{1}{\sqrt{a_i} + \sqrt{a_{i+1}}} = \frac{2020}{\sqrt{a_1} + \sqrt{a_{2021}}}$	(i) If $b_i > 0, \forall i \in \mathbb{N}$ then $\log b_1, \log b_2 \dots \log b_n$ is an A.P.	(P) If $c_i \neq 0, i \in \mathbb{N}$ then $\frac{1}{c_1}, \frac{1}{c_2}, \dots, \frac{1}{c_n}$ are always in A.P.
(II) If $a_i > 0 \forall i \in \mathbb{N}$ then $\sum_{i=1}^{2020} \frac{1}{\sqrt{a_i} + \sqrt{a_{i+1}}} = \frac{2020}{\sqrt{a_1} + \sqrt{a_{2020}}}$	(ii) If $b_1 = 2, b_2 = 1$ then $\sum_{i=1}^{\infty} b_i = 4$	(Q) $c_1^2, c_2^2, \dots, c_n^2$ are always in G.P.
(III) If 'd' is common difference of $a_1, a_2, \dots, a_{2020}$ then $\sum_{r=1}^{2020} \frac{1}{a_r a_{r+1}} = \frac{2020}{a_1 a_{2021}}$	(iii) If $b_1 = 1,$ $b_3 = \frac{1}{4}$ and $b_2 > 0$ then $\sum_{i=1}^{\infty} b_i = \frac{4}{3}$	(R) $c_1 \neq c_2 \neq c_3$ then $\frac{c_1}{c_1 - c_2} = \frac{c_1 + c_3}{c_1 - c_3}$
(IV) If $a_i > 0 \forall i \in \mathbb{N}$ then $\sum_{i=1}^{2019} \frac{1}{\sqrt{a_i} + \sqrt{a_{i+1}}} = \frac{2019}{\sqrt{a_1} + \sqrt{a_{2020}}}$	(iv) $\sum_{i=1}^n b_i = \frac{b_2(b_1^n - b_2^n)}{b_1^{n-1}(b_2 - b_3)}$	(S) $c_1 = 1, c_2 = 3$ then $c_3 = -3$

11. Which of the following options is the only **CORRECT** combination ?
 (A) (I) (i) (P) (B) (I) (ii) (Q) (C) (I) (iii) (S) (D) (I) (iv) (Q)
12. Which of the following options is the only **CORRECT** combination ?
 (A) (II) (ii) (Q) (B) (III) (ii) (P) (C) (II) (ii) (P) (D) (I) (iii) (Q)
13. Which of the following options is the only **INCORRECT** combination ?
 (A) (III) (iv) (S) (B) (IV) (ii) (P) (C) (IV) (iii) (R) (D) (IV) (iv) (P)

Space for Rough Work

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If $x_1, x_2, x_3, \dots, x_{10}$ lying in $\left[0, \frac{\pi}{2}\right]$ such that arithmetic mean of $\cos x_1, \cos x_2, \dots, \cos x_{10}$ is $|\operatorname{cosec} \alpha|$, then harmonic mean of $\cos x_1, \cos x_2, \dots, \cos x_{10}$ is $(\alpha \neq n\pi, n \in \text{Integer})$
2. If $P = \operatorname{cosec} \frac{\pi}{8} + \operatorname{cosec} \frac{2\pi}{8} + \operatorname{cosec} \frac{3\pi}{8} + \operatorname{cosec} \frac{13\pi}{8} + \operatorname{cosec} \frac{14\pi}{8} + \operatorname{cosec} \frac{15\pi}{8}$
 & $Q = \prod_{r=1}^{10} \log(\sin(10r)^\circ)$, then value of $P + Q$ is

Space for Rough Work

3. 3 natural numbers x, y, z are in G.P. & x, y^2, z are in A.P., then the minimum value of $x + 2y + z$ is equal to
4. If $x, y \in \mathbb{R}^+$ and satisfying $x + y = 3$ then the maximum value of x^2y is
5. The least value of the expression $2\log_5 x - \log_x(0.04)$, for $x > 1$, is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	7	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	28
I(ii)	Matching Lists Type (Single correct option)	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18
II	Numerical Value Type (Up to second decimal place)	5	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	15

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator